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MODERN INSTITUTE OF TECHNOLOGY AND RESEARCH CENTRE, ALWAR

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Subject & Subject Code: 2FY1-02 Engineering Physics
Semester: II **Session:** 2022-23 (Even Sem.)
Branch: AIDS, ME, CV, EE **Batch:** C & D
Unit No.:2nd **Date of Submission:** 20/03/2023

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References:

1. A. Beiser, "Concept of Modern Physics", Publisher, Tata Mc Graw Hill, Fifth Edition.
2. Oscillation and Waves, N. K. Bajaj

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LECTURE -8

Topic: - Introduction to quantum Mechanics

INTRODUCTION

Quantum mechanics is an important intellectual achievement of the 20th century. It is one of the more sophisticated fields in physics that has affected our understanding of nano-meter length scale systems important for chemistry, materials, optics, and electronics. The existence of orbitals and energy levels in atoms can only be explained by quantum mechanics. Quantum mechanics can explain the behaviors of insulators, conductors, semi-conductors, and giant magneto-resistance. It can explain the quantization of light and its particle nature in addition to its wave nature. Quantum mechanics can also explain the radiation of hot body, and its change of color with respect to temperature. It explains the presence of holes and the transport of holes and electrons in electronic devices.

Quantum mechanics has played an important role in photonics, quantum electronics, and micro-electronics. But many more emerging technologies require the understanding of quantum mechanics; and hence, it is important that scientists and engineers understand quantum mechanics better. One area is nano-technologies due to the recent advent of nanofabrication techniques. Consequently, nano-meter size systems are more common place. In electronics, as transistor devices become smaller, how the electrons move through the device is quite different from when the devices are bigger: nano-electronic transport is quite different from micro-electronic transport.

The quantization of electromagnetic field is important in the area of nano-optics and quantum optics. It explains how photons interact with atomic systems or materials. It also allows the use of electromagnetic or optical field to carry quantum information. Moreover, quantum mechanics is also needed to understand the interaction of photons with materials in solar cells, as well as many topics in material science.

When two objects are placed close together, they experience a force called the Casimir force that can only be explained by quantum mechanics. This is important for the understanding of micro/nano-electromechanical sensor systems (M/NEMS). Moreover, the understanding of spins is important in spintronics, another emerging technology where giant magneto-resistance, tunneling magneto-resistance, and spin transfer torque are being used. Quantum mechanics is also giving rise to the areas of quantum information, quantum communication, quantum cryptography, and quantum computing. It is seen that the richness of quantum physics will greatly affect the future generation technologies in many aspects.

QUANTUM MECHANICS IS BIZARRE

The development of quantum mechanics is a great intellectual achievement, but at the same time, it is bizarre. The reason is that quantum mechanics is quite different from classical physics. The development of quantum mechanics is likened to watching two players having a game of chess, but the watchers have not a clue as to what the rules of the game are. By observations, and conjectures, finally the rules of the game are outlined. Often, equations are conjectured like conjurors pulling tricks out of a hat to match experimental observations. It is the interpretations of these equations that can be quite bizarre.

Quantum mechanics equations were postulated to explain experimental observations, but the deeper meanings of the equations often confused even the most gifted. Even though Einstein received the Nobel prize for his work on the photo-electric effect that concerned that light energy is quantized, he himself was not totally at ease with the development of quantum mechanics as charted by the younger physicists. He was never comfortable with the probabilistic interpretation of quantum mechanics by Born and the Heisenberg uncertainty principle: “God doesn't play dice," was his statement assailing the probabilistic interpretation. He proposed “hidden variables" to explain the random nature of many experimental observations. He was thought of as the “old fool" by the younger physicists during his time. Schrodinger came up with the bizarre Schrodinger cat paradox" that showed the struggle that physicists had with quantum mechanics's interpretation. But with today's understanding of quantum mechanics, the paradox is a thing of yesteryear.

The latest twist to the interpretation in quantum mechanics is the parallel universe view that explains the multitude of outcomes of the prediction of quantum mechanics. All outcomes are possible, but with each outcome occurring in different universes that exist in parallel with respect to each other.

THE WAVE NATURE OF A PARTICLE WAVE PARTICLE DUALITY

The quantized nature of the energy of light was rest proposed by Planck in 1900 to successfully explain the black body radiation. Einstein's explanation of the photoelectric effect further asserts the quantized nature of light, or light as a photon.1 However, it is well known that light is a wave since it can be shown to interfere as waves in the Newton ring experiment as far back as 1717.

The wave nature of an electron is revealed by the fact that when electrons pass through a crystal, they produce a diffraction pattern. That can only be explained by the wave nature of an electron. This experiment was done by Davisson and Germer in 1927. De Broglie hypothesized that the wavelength of an electron, when it behaves like a wave, is

$$\lambda = \frac{h}{p} \qquad (1.3.1)$$

where h is the Planck's constant, p is the electron momentum,2 and

$$h \approx 6.626 \times 10^{-34}\,\text{Joule} \cdot \text{second} \qquad (1.3.2)$$

When an electron manifests as a wave, it is described by

$$\psi(z) \propto \exp(ikz) \qquad (1.3.3)$$

where $k = 2\pi/\lambda$. Such a wave is a solution to[3]

$$\frac{\partial^2}{\partial z^2}\psi = -k^2\psi \tag{1.3.4}$$

A generalization of this to three dimensions yields

$$\nabla^2\psi(\mathbf{r}) = -k^2\psi(\mathbf{r}) \tag{1.3.5}$$

We can define

$$p = \hbar k \tag{1.3.6}$$

where $\hbar = h/(2\pi)$.[4] Consequently, we arrive at an equation

$$-\frac{\hbar^2}{2m_0}\nabla^2\psi(\mathbf{r}) = \frac{p^2}{2m_0}\psi(\mathbf{r}) \tag{1.3.7}$$

where

$$m_0 \approx 9.11 \times 10^{-31}\text{kg} \tag{1.3.8}$$

The expression $p^2/(2m_0)$ is the kinetic energy of an electron. Hence, the above can be considered an energy conservation equation.

The Schrödinger equation is motivated by further energy balance that total energy is equal to the sum of potential energy and kinetic energy. Defining the potential energy to be $V(\mathbf{r})$, the energy balance equation becomes

$$\left[-\frac{\hbar^2}{2m_0}\nabla^2 + V(\mathbf{r})\right]\psi(\mathbf{r}) = E\psi(\mathbf{r}) \tag{1.3.9}$$

Where E is the total energy of the system. The above is the time-independent Schrodinger equation. The ad hoc manner at which the above equation is arrived at usually bothers a beginner in the field. However, it predicts many experimental outcomes, as well as predicting the existence of electron orbital’s inside an atom, and how electron would interact with other particles.

LECTURE -9

Topic: - Wave-particle duality

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(1) state de-broglie hypothesis. Using de-broglie wavelength expression, show that an Electron accelerated by a potential difference v volt is

$$\lambda = \frac{1.226x10-9}{\sqrt{V}}$$

Ans .

According to de-Broglie, in the universe, whole of energy appears in the form of radiation and matter. Since nature loves symmetry, if radiation which normally behaves as a wave can behave as a particle, then one can even expect that, entities like electrons, protons etc, which ordinarily behave as particle also exhibits properties of waves under appropriate circumstances. These **waves associated with the matter are known as matter waves.** The waves associated with the moving particle are called matter wave or de-Broglie wave. The wavelength associated with particles with mass 'm' and moving with certain velocity 'v' and momentum 'p' is given by,

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Consider an electron of mass 'm' which is at rest and is subjected to a potential difference of 'V' volt. The electrical energy of the electron (eV) will appear as kinetic energy of the electron.

i.e

$$ie., \quad E = eV$$

$$and \quad E = \frac{1}{2}mv^2$$

$$\therefore \quad m^2v^2 = 2mE$$

$$\Rightarrow \quad mv = p = \sqrt{2mE}$$

Wavelength of electron wave $\lambda = \frac{h}{p} = \frac{h}{mv}$

$$\therefore \lambda = \frac{h}{\sqrt{2mE}}$$

But $E = eV$

$$\boxed{\therefore \lambda = \frac{h}{\sqrt{2meV}}}$$

By substituting the values of h- Planck's constant, e- electron charge and m- mass of electron, $h = 6.626 \times 10^{-34}$ joule –sec , $e = 1.6 \times 10^{-19}$ coulomb, $m = 9.1 \times 10^{-31}$ kg

We get, $\lambda = \frac{1.226 \times 10^{-9}}{\sqrt{V}}$

Example 1 :- find De –Brogli wavelength for an electron accelerated by a voltage 20 V.

Ans .

$$\lambda = \frac{1.226 \times 10^{-9}}{\sqrt{V}} = \frac{1.226 \times 10^{-9}}{\sqrt{20}} = 2.7 \times 10^{-10} \text{ m} = 2.7 \text{ A}^\circ$$

(2) Define group velocity and obtain the expression for the same.

The velocity with which the wave packet- which is formed due to the superposition of two or more waves of slightly different wavelengths is transmitted is called as group velocity. (or) It is the velocity with which resultant wave packet formed due to the superposition of two or more waves propagates.

Expression for group velocity:

Let us consider two traveling waves of same amplitude but of slightly different wavelengths and frequencies. Let them be represented by the equations

$Y1 = A \sin(\omega t - kx)$ .... (1)

$Y2 = A \sin\{(\omega + \Delta\omega)t - (k + \Delta k)x\}$ ....(2)

Where, y1 and y2 are the displacements along y-direction ω and $\omega + \Delta\omega$, k and $k + \Delta k$ are the angular frequency and propagation constants of the two waves. $\Delta\omega$ and Δk represents the small changes in angular frequency and propagation constant.

The resultant of the two waves due to the superposition of the waves is given by,

$$\begin{aligned} Y &= y_1 + y_2 \\ &= A\sin(\omega t - kx) + A\sin\{(\omega + \Delta\omega)t - (k + \Delta k)x\} \end{aligned}$$

Using the identity, $\sin a + \sin b = 2\cos\left(\frac{a-b}{2}\right)\sin\left(\frac{a+b}{2}\right)$ we can write

$$Y = 2A\cos\left\{\left(\frac{\Delta\varpi}{2}\right)t - \left(\frac{\Delta k}{2}\right)x\right\} \cdot \sin(\varpi t - kx)$$

This eqn canbewritten as

$$Y = R\sin(\varpi t - kx) - u - - - -(3)$$
$$\text{Where, } R = 2A\cos\left\{\left(\frac{\Delta\sigma}{2}\right)t - \left(\frac{\Delta k}{2}\right)x\right\}$$

' R' is the amplitude of the resultant wave. From this expression it is clear that 'R' is not a constant. Equation (3) represents wave equation of the wave packet. This also represents the equation of matter wave. Therefore comparing with standard wave equation we can write the velocity of the wave group as,

$$V_g = (\frac{\Delta\varpi/2}{\Delta k/2}) = (\frac{\Delta\varpi}{\Delta k})$$

i.e $$V_g = (\frac{\Delta\varpi}{\Delta k})$$

By applying limit we can write ,

$$V_g = (\frac{\mathrm{d}\varpi}{\mathrm{d}k})$$

This is the expression for group velocity.

LECTURE -10

Topic: - Matter waves, Wave function and basic postulates

WHAT ARE MATTER WAVES?

The waves associated with a moving particle are called matter waves or de-Broglie waves. The de-Broglie wavelength associated with a particle with mass 'm' and moving with certain velocity 'v' and momentum 'p' is given by,

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Properties of matter waves:

1. Matter waves are the waves associated with matter in motion. Wavelength of the matter waves is given by λ = h/p.

2. Matter waves are not progressive waves. They are localized waves.

3. Matter waves are not single waves. They are group of waves (wave packet) assumed formed due to the superposition of two or more progressive waves.

4. Matter waves called pilot waves and they represent the direction of propagation of matter.

5. Unlike electromagnetic waves, matter waves will not have constant speed. The speed of matter waves depends on the mass of the particle.

6. $V_g \times V_p = C^2$ shows that Vp, phase velocity of matter waves is greater than velocity of light. This shows that matter waves not physical waves.

WHAT IS A WAVE FUNCTION? EXPLAIN THE PROPERTIES OF WAVE FUNCTION.

Ans: The quantity that characterizes the de–Broglie wave or matter wave is called the wave function. It is usually denoted as $\Psi = (x, y, z, t)$. This gives complete information about the state of a physical system at a particular time. It is also called the state function and represents the probability amplitude. If 'Ψ' is large, the probability of finding the particle is also large and if 'Ψ' is small then the probability of finding the particle is small. The wave function gives the likelihood of finding the particle at a given instant and at a given position inside the wave packet.

Properties of wave functions:

There are certain properties that an acceptable wave function 'Ψ' must satisfy:

1. In order to avoid infinite probabilities, Ψ must be finite for all values of x, y, z.

2. In order to avoid multiple values of the probability, Ψ must be single valued. i.e., for each set of x, y and z, Ψ must have a unique value.

3. For finite potentials, Ψ and ∂Ψ/∂*x*, ∂Ψ/∂*y*, ∂Ψ/∂*z*, must be continuous in all regions.

4. In order to normalize the wave function, Ψ must approach to zero as 'x' approaches to (+/-) infinity.

WRITE THE PHYSICAL SIGNIFICANCE OF WAVE FUNCTION (OR) MAX BORN'S INTERPRETATION OF WAVE FUNCTION.

Ans:

In any physical wave if 'A' is the amplitude of the wave, then the energy density i.e., energy per unit volume is equal to 'A2'. Similar interpretation can be made in case of mater wave also. In matter wave, if 'Ψ 'is the wave function of matter waves at any point in space, then the particle density at that point may be taken as proportional to'Ψ'. **Thus Ψ2 is a measure of particle density**.

According to Max Born $\Psi\Psi^* = \Psi^2$ gives the probability of finding the particle in the state 'Ψ'. i.e., $\lfloor \Psi \rfloor^2$ is a measure of probability density. The probability of finding the particle in a volume *dv* (= *dx dy dz*) is given by $\lfloor \Psi \rfloor^2 dv$ ***or*** *dx.dy.dz*

Since the particle has to be present somewhere, total probability of finding the particle somewhere is unity i.e., particle is certainly to be found somewhere in space.

$$\text{i.e.,} \quad \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} |\Psi|^2 \, dx.dy.dz = 1 \quad \text{(or)} \quad \int_{-\infty}^{+\infty} |\Psi|^2 \, dv = 1$$

This condition is called **Normalization condition.** A wave function which satisfies this condition is known as normalized wave function.

STATE AND EXPLAIN HEISENBERG'S UNCERTAINTY PRINCIPLE. GIVE ITS PHYSICAL SIGNIFICANCE.

Statement: It is impossible to determine simultaneously both the position and momentum of a quantum mechanical particle moving inside the wave packet with equal accuracy. In any simultaneous determination of the position and momentum of a particle, the product of the corresponding uncertainties inherently present in the measurement is equal to, or greater than $\frac{(h)}{(4\pi)}$.

Explanation: Consider a quantum particle moving inside a wave packet of width Δx. Then inside the wave packet it is not possible to find the exact position of the particle and any attempt to measure it will have an uncertainty (error) which less than or equal to Δx. The maximum uncertainty involved in measuring the position of the particle within the wave packet is Δx. Since position cannot be measured accurately, there will be an uncertainty in measuring momentum also. According to Heisenberg's uncertainty principle, the product of uncertainty involved in the measurement of these two quantities is given by the relation,

i.e., $$\Delta x.\Delta p \geq h/4\pi.$$

Where, Δx ---- is uncertainty in position & Δp ----- is uncertainty in momentum. The problem of uncertainty is a logical consequence of **dual nature of matter. It can be applied to any conjugate physical quantities** such as energy and time, angular position and angular momentum etc.

Other forms of Uncertainties are,

$$\Delta E.\Delta t \geq \frac{h}{4\pi}$$

$$\Delta L.\Delta \theta \geq \frac{h}{4\pi}$$

Where, ΔE, Δt are the uncertainties in the measurement of energy and time, and ΔL, $\Delta\theta$ are the uncertainties in the measurement of angular momentum and angular position.

Physical significance of Heisenberg's uncertainty principle:

According to Heisenberg's uncertainty principle it is not possible to locate exactly the position of a quantum mechanical particle as well as not possible to measure accurately the momentum of the particle. Any attempt in measuring these two quantities will leads in to an uncertainty of **h/4π**. Therefore one has to think of finding the most probable values for position of the particle and measuring the most probable value for momentum of the particle. Such probable values can be found by using mathematical function called probability density function denoted by the symbol '**Ψ'.**

LECTURE -11

TOPIC :-Time dependent Schrodinger's Wave Equation

TIME DEPENDENT SCHROEDINGER WAVE EQUATION

The quantity that characterises the matter waves (De Broglie waves) is called the wave function (ψ). . It may be a complex function of space coordinate and time. Let us assume a wave function that represents a particle moving in the +x direction with momentum p and total energy E as follows

$$\psi = Ae^{-i\omega(t-\frac{x}{v})} = Ae^{-2\pi i(vt-\frac{x}{\lambda})}$$

$$\psi = Ae^{-\frac{2\pi i}{h}(Et-px)} \text{------------------ (4.78)}$$

Where $E = hv$ and $p = \frac{h}{\lambda}$

Differentiating equation (4.78) twice w.r.t to x , we get

$$\frac{\partial^2\psi}{\partial x^2} = -\frac{4\pi^2p^2}{h^2} \cdot Ae^{-\frac{2\pi i}{h}(Et-px)}$$

$$\frac{\partial^2\psi}{\partial x^2} = -\frac{4\pi^2 p^2}{h^2}\psi$$

$$\Rightarrow p^2\psi = -\frac{h^2}{4\pi^2}\frac{\partial^2\psi}{\partial x^2} \quad ---------------(4.79)$$

Differentiating equation (4.78) twice w.r.t to t , we get

$$\frac{\partial\psi,}{\partial t} = \frac{-2\pi Ei}{h} \cdot Ae^{-\frac{2\pi i}{h}(Et-px)}$$

$$\frac{\partial\psi,}{\partial t} = \frac{-2\pi Ei}{h} \cdot \psi$$

$$\mathrm{E}\,\psi = -\frac{\mathrm{h}}{2\pi i}\frac{\partial\psi,}{\partial t} \text{ ----------------------- } (4.80)$$

Let us assume that motion of particle in non-relativistic limits, i.e., the speed of particle is much small compared with that of light, then the total energy E of this particle is the sum of its kinetic energy $\frac{p^2}{2m}$ and its potential energy V.

$$\mathrm{E} = \frac{P^2}{2m} + \mathrm{V}$$

By acting this energy operator on wave function ψ we get

$$E\psi = \frac{P^2\psi}{2m} + V\psi$$

Substituting the value of $E\psi$ and $P^2\psi$ from equation (4.62) and equation (4.80) we get

$$-\frac{h}{2\pi i}\frac{\partial\psi}{\partial t} = -\frac{h^2}{2m \times 4\pi^2}\frac{\partial^2\psi}{\partial x^2} + V\psi$$

$\frac{ih}{2\pi}\frac{\partial\psi}{\partial t} = -\frac{h^2}{8\pi^2 m}\frac{\partial^2\psi}{\partial x^2} + V\psi$ ---------------------------------------(4.82)

This equation is known as **time dependent Schroedinger equation** in one dimension, because this equation contains the time (t).

In three dimensions, the time dependent Schroedinger equation is

$-\frac{h^2}{8\tau^2 m}(\frac{\partial^2\psi}{\partial \mathrm{x}^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2}) + V\psi = \frac{ih}{2\pi}\frac{\partial\psi}{\partial t}$ ------------------------------(4.83)(a)

$-\frac{h^2}{8\pi^2 m}\nabla^2\psi + V\psi = \frac{ih}{2\pi}\frac{\partial\psi}{\partial t}$ ------------------------------------(4.83)(b)

Any restriction on particle's motion will affect the potential energy V, which, in general, is a function of space (x, y, z) and time t . Once V is known, the equation may be solved for ψ

This equation is used in explaining non-stationary phenomenon e.g. an electric transition between two states of an atom equation (4.83) (b) can be written as

$(-\frac{h^2}{8\pi^2 m}\nabla^2 + V)\psi = \frac{ih}{2\pi}\frac{\partial}{\partial t}\psi$

$$\Rightarrow H\psi = E\psi$$

Where $H = -\frac{h^2}{8\pi^2 m}\nabla^2 + V$; is called **Hamiltonians operator**

and

$E = \frac{ih}{2\pi}\frac{\partial}{\partial t}$; is the **energy operator**

LECTURE -12

Topic :- Time independent Schrodinger's Wave Equation

DISCUSS THE SOLUTION FOR ENERGY EIGEN VALUES OF A PARTICLE FREELY MOVING IN SPACE. OR SOLVE SCHRODINGER WAVE EQUATION FOR ALLOWED ENERGY VALUES IN THE CASE OF A PARTICLE MOVING IN A FREE SPACE.

Ans: A free particle means, it is not under the influence of any kind of field or force. Thus it has zero potential, i.e., V = 0. Hence Schrodinger's equation becomes,

$$\frac{d^2\Psi}{dx^2} + \frac{8\pi^2 m}{h^2} E\Psi = 0$$

Solution for this equation is

$$\Psi = A\cos(kx) + B\sin(kx)$$

Since there are no boundaries for the particle, the constants A, B & k cannot be determined and these constants can have any values.

We can show that the energy Eigen equation for the particle moving in space is given by

$$E = \frac{n^2 h^2}{8ma^2} \qquad n = \frac{2a}{h}\sqrt{2Em}$$

The value of 'n' which depends only on 'a', and for a free particle width of potential well'a' is infinite. As 'a' tends to infinity, the value of 'n' can also tend to have infinite values. i.e., 'n ' can take all possible values up to infinity. Hence energy of the free particle continuous. In other words energy is not quantized or the free particle is a classical particle.

OBTAIN THE TIME INDEPENDENT SCHRODINGER WAVE EQUATION.

Ans: Schrodinger wave equation is a differential equation which plays the same role as that of Newton's equations of motions. It is used to describe the motion of particle inside a wave packet.

According to de-Broglie theory, for a particle of mass 'm', moving with a velocity'v', the wavelength associated with it is,

$$\lambda = \frac{h}{p}$$

The wave equation for a de-Broglie wave can be written in complex notation as,

$$\Psi = Ae^{i(\omega t - kx)} \ldots\ldots\ldots\ldots.(1)$$

where, A is the amplitude, ω is angular frequency and k is the wave vector.

Differentiate equation (1) with respect to't' twice, we get

$$\frac{d^2\Psi}{dt^2} = -\omega^2\left[Ae^{i(\omega t - kx)}\right]$$

(or)

$$\frac{d^2\Psi}{dt^2} = -\omega^2\Psi \ldots\ldots\ldots.(2)$$

we have differential equation for the traveling wave as,

$$\frac{d^2y}{dx^2} = \frac{1}{v^2}\frac{d^2y}{dt^2} \ldots\ldots\ldots.(3)$$

where, y is displacement and 'v' is velocity of wave By analogy, we can write the wave equation for de-Broglie wave associated with the motion of a free particle as,

$$\frac{d^2\Psi}{dx^2} = \frac{1}{v^2}\frac{d^2\Psi}{dt^2} \text{..........(4)}$$

This represents the de-Broglie wave propagating along x-direction with a velocity 'v' and 'Ψ 'is the displacement. From equation (2) and (4),

$$\frac{d^2\Psi}{dx^2} = -\frac{1}{v^2}\omega^2\Psi$$

$$substituting\ \ v = \nu\lambda \quad and \quad \varpi = 2\pi\nu$$

$$\frac{d^2\Psi}{dx^2} = -\frac{1}{(\nu\lambda)^2}(2\pi)^2\nu^2\Psi$$

$$\frac{d^2\Psi}{dx^2} = -\frac{4\pi^2}{\lambda^2}\Psi$$

$$\Rightarrow \frac{1}{\lambda^2} = -\frac{1}{4\pi^2\Psi}\frac{d^2\Psi}{dx^2} \text{............(5)}$$

The kinetic energy of a moving particle of mass 'm' and velocity 'v' is given by

$$\frac{1}{2}mv^2 = \frac{m^2v^2}{2m} = \frac{p^2}{2m}$$

But we have from equation (1), p = (h/λ)

$$K.E. = \frac{h^2}{2m}\frac{1}{\lambda^2}$$

Substitute for $(1/\lambda)^2$ from equation (5).

$$K.E = \frac{-h^2}{2m}\frac{1}{4\pi^2\Psi}\frac{d^2\Psi}{dx^2} \text{...............(6)}$$

Let there be a field where the particle is present. Depending on its position in the field, the particle will possess certain potential energy. Then we can write

Total energy = Kinetic energy + Potential energy

$$E = \frac{p^2}{2m} + V$$

$$\frac{p^2}{2m} = E - V$$

From equation (6), we can write

$$\frac{-h^2}{8m\pi^2}\frac{1}{\Psi}\frac{d^2\Psi}{dx^2} = (E - V)$$

$$\therefore \frac{d^2\Psi}{dx^2} + \frac{8\pi^2 m}{h^2}(E - V)\Psi = 0$$

This is the time independent Schrodinger's wave equation in one-dimension. It is generally called "steady state schroedinger equaction" for a particle of mass m with total energy E and potential energy V, moving in x direction. In three dimension space , the time independent schroedinger equaction

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0$$

$$\nabla^2\psi + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0$$

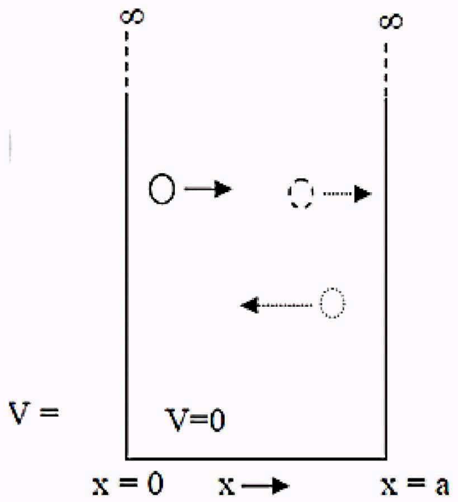
where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is the laplacian operator

LECTURE -13

Topic: - Applications of the Schrodinger's Equation

DETERMINE ENERGY EIGEN VALUES AND EIGEN FUNCTIONS FOR A PARTICLE IN A ONE DIMENSIONAL POTENTIAL WELL/ BOX OF INFINITE HEIGHT.

Ans: Consider a particle of mass ‘m’ moving inside a one dimensional potential well of infinite height and finite width ‘a’. Assume potential inside the well as constant. Outside this region potential energy ‘V’ is infinity.



As a result the particle has to move only inside the well like a free particle. Inside the box, the Schrodinger’s equation is given by,

$$\nabla^2\psi + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0 \quad ------------(1)$$

By taking potential V= 0 (which is constant), we can write equ. (1) as

$$\nabla^2\psi + \frac{8\pi^2 m}{h^2}(E)\psi = 0$$

$$\nabla^2\psi + k^2\psi = 0 \quad ------------------------------------(2)$$

Where

$$k^2 = \frac{8\pi^2 m}{h^2}(E) \quad -----------------(2a)$$

Discussion of the solution:

The solution of the above equation (2) is given by

Ψ = *A*cos *kx* + *B*sin *kx*..........(3)

Where A & B are constants depending on the boundary condition. Now apply boundary conditions for this,

Condition: I

at x = 0, ψ = 0.

From equation (3) we get A = 0.

Condition: II

at x = a, ψ = 0

From equation (3) we get, 0 = B sin(ka)

Since B ≠ 0

$$\sin ka = 0$$

$$\Rightarrow ka = n\pi$$

$$\therefore k = \frac{n\pi}{a} \quad \text{where, n = 1,2,3................}$$

$$k^2 = \frac{n^2\pi^2}{a^2}$$

Therefore the wave function Ψ becomes

$$\Psi = B\sin\left(\frac{n\pi}{a}\right)x \;.................\; (3a)$$

Substitute the value of k^2 in equation (2a) we get,

$$\frac{8m\pi^2 E}{h^2} = \frac{n^2\pi^2}{a^2}$$

$$E = \frac{n^2h^2}{8ma^2} \text{................}(4)$$

This is called energy Eigen equation for the particle in one dimension box.

When, n = 0, $\psi_n = 0$. This means to say that the electron is not present inside the box which is not true. Hence the lowest value of 'n' is 1.

∴ The lowest energy corresponds to 'n' which is acceptable is for n =1 and the corresponding energy is called the **zero-point energy** or **ground state energy** and all the other states which corresponds to **n >1 are called excited states.**

$$E_{zero\text{-}point} = E_0 = \frac{h^2}{8ma^2} \text{ -----------------}(5)$$

Energy in the first excited state is given by (n=2)

$$E_1 = \frac{4h^2}{8ma^2} \text{ ------------------------------}(6)$$

Energy in the 2nd excited state is given by (n=3)

$$E_2 = \frac{9h^2}{8ma^2} \text{ ------------------------------}(7)$$

Equation (5), (6) & (7) represent the energy Eigen values of the particle moving inside a one dimensional potential well.

To evaluate B in equation (3), one has to perform normalization of wave function.

Normalization:

Consider the equation,

$$\Psi = B\sin\left(\frac{n\pi}{a}\right)x$$

The integral of the wave function over the entire space in the box must be equal to unity because there is only one particle within the box, the probability of finding the particle

is 1. i.e. $\int_0^a |\Psi|^2 dx = 1$

$$\int_0^a B^2 \sin^2\left(\frac{n\pi}{a}\right)x \ dx = 1 \text{ , but } \sin^2\theta = \frac{1}{2}(1-\cos 2\theta)$$

$$\frac{B^2}{2}\int_0^a \left(1-\cos\frac{2n\pi x}{a}\right)dx = 1$$

$$\therefore \frac{B^2 a}{2} = 1$$

$$\Rightarrow B = \sqrt{2/a}$$

Thus, the normalized wave function of a particle in a one-dimensional box is given by,

$$\Psi_n = \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}\right)x \quad \text{where n=1,2,3}$$

Since the particle in a box is a quantum mechanical problem we need to evaluate the most probable location of the particle in a box and its energies at different permitted state.

Let us consider first three cases

Case (1) : n =1.

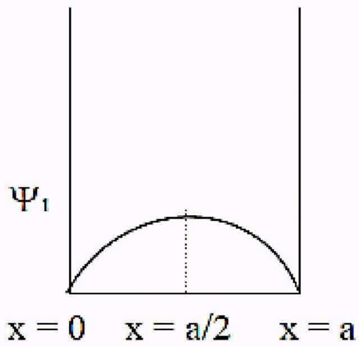
This is the ground state and the particle is normally found in this state.

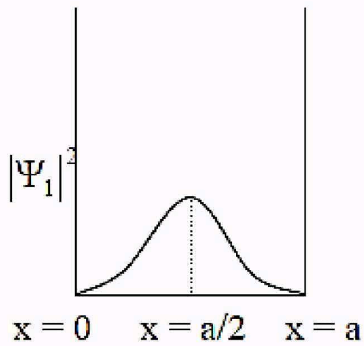
For n =1, the Eigen function is

$$\Psi_1 = B\sin\left(\frac{\pi}{a}\right)x$$

In the above equation $\Psi = 0$ for both $x = 0$ & $x = a$. but Ψ_1 has maximum value for $x = a/2$.

And $|\Psi_1|^2 = 0$ at $x = 0$ and $x = a$, and $|\Psi_1|^2$ is maximum at $x = (a/2)$.



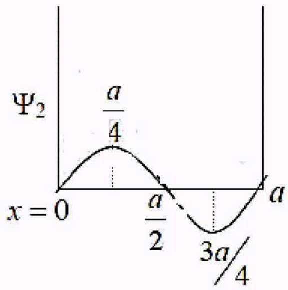


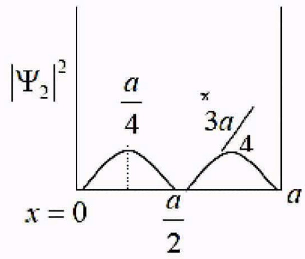
A plot of Ψ_1 and Ψ_1^2 - the probability density, versus 'x' is as shown. From the figure, it is clear that at ground state the probability of finding the particle is maximum at the centre and the particle cannot be found at the walls of the potential well.

Case (2): n =2

This is the first excited state. The Eigen function for this state is given by

$$\Psi_2 = B\sin\left(\frac{2\pi}{a}\right)x$$





now, Ψ_2 =0 for the values $x = 0,\ \frac{a}{2},\ a$

and Ψ_2 reaches maximum at $x = \frac{a}{4},\ \frac{3a}{4}$

these facts are seen in the following plot.

From the figure it can be seen that $|\Psi_1|^2 = 0$ at $x = 0,\ a/2, a$. It means that in the first excited state the particle cannot be observed either at the walls or at the center and the probability of

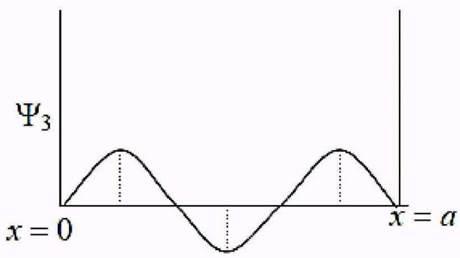
Case (3): n =3.

This is the second excited state and the Eigen function for this state is given by

$$\Psi_3 = B\sin\left(\frac{3\pi}{a}\right)x$$

now, Ψ_3 =0 for the values $x = 0,\ \frac{a}{3},\ \frac{2a}{3},\ a$

and Ψ_2 reaches maximum at $x = \frac{a}{6},\ \frac{a}{2},\ \frac{5a}{6}$



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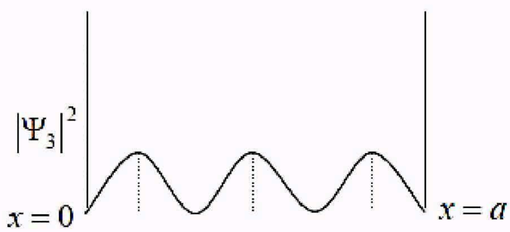
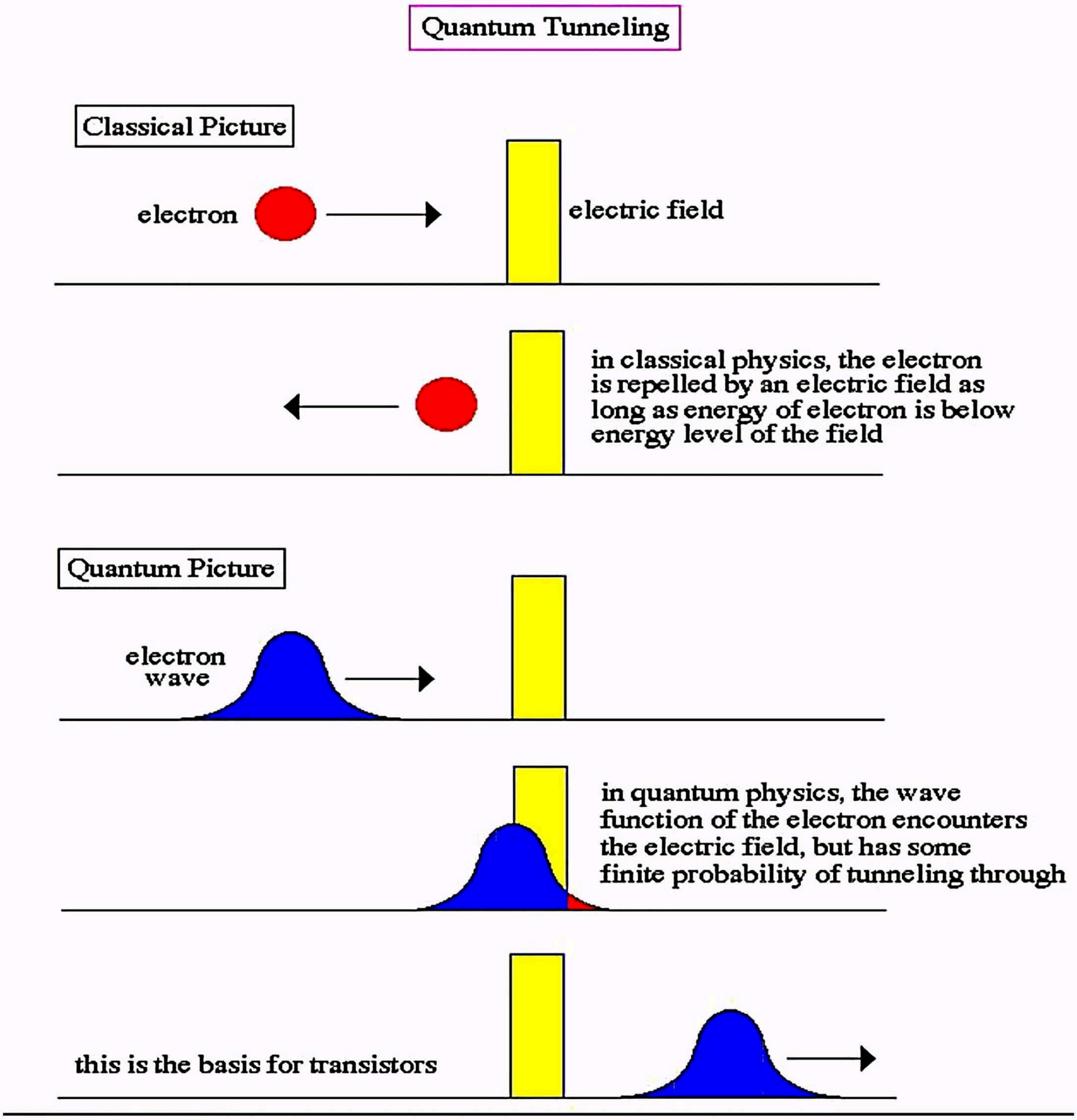


The plot of $|\Psi_3|^2$ versus 'x' has maxima at $x = a/6, a/2, 5a/6$ at which the particle is most likely to be found.

LECTURE -14

Quantum Tunneling

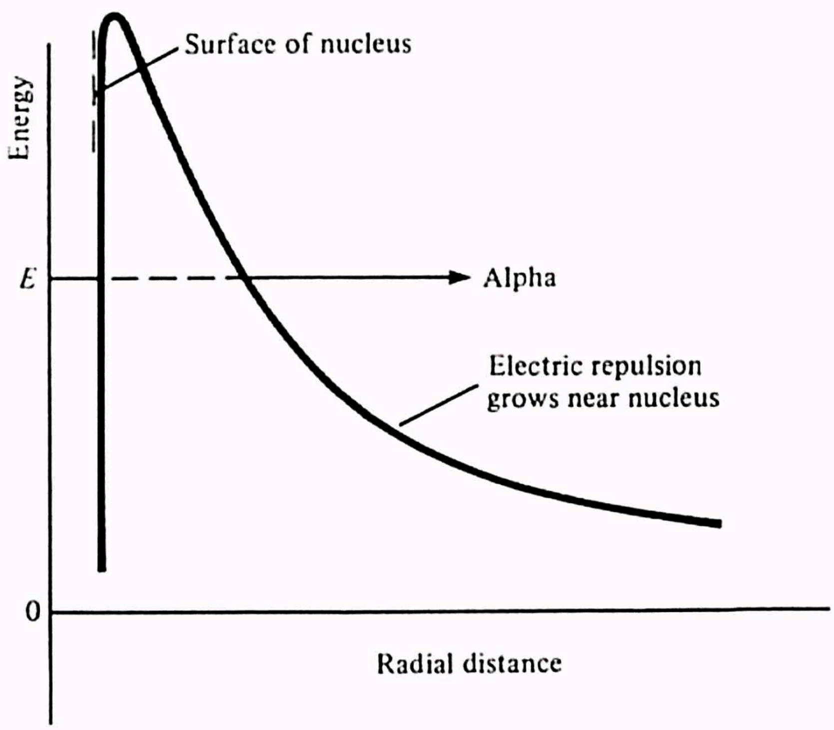


Quantum Tunneling :

The phenomenon of tunneling, which has no counterpart in classical physics, is an important consequence of quantum mechanics. Consider a particle with energy E in the inner region of a one-dimensional potential well V(x). (A potential well is a potential that has a lower value in a certain region of space than in the neighboring regions.) In classical mechanics, if E < V (the maximum height of the potential barrier), the particle remains in the well forever; if E > V, the particle escapes. In quantum mechanics, the situation is not so simple. The particle can escape even if its energy E is below the height of the barrier V, although the probability of escape is small unless E is close to V. In that case, the particle may tunnel through the potential barrier and emerge with the same energy E.

The phenomenon of tunneling has many important applications. For example, it describes a type of radioactive decay in which a nucleus emits an alpha particle (a helium nucleus). According to the quantum explanation given independently by George Gamow and by Ronald W. Gurney and Edward Condon in 1928, the alpha particle is confined before the decay by a potential. For a given nuclear species, it is possible to measure the energy E of the emitted alpha particle and the average lifetime of the nucleus before decay. The lifetime of the nucleus is a measure of the probability of tunneling through the barrier--the shorter the lifetime, the higher the probability.

Potential barrier around a uranium nucleus presented to an alpha particle. The central well is due to the average nuclear attraction of all the nucleons and the hill is due to the electric repulsion of the protons. Alpha particles with energy *E* trapped inside the nuclear well may still escape to become alpha rays, by quantum mechanically tunnelling through the barrier.



With plausible assumptions about the general form of the potential function, it is possible to calculate a relationship between and E that is applicable to all alpha emitters. This theory, which is borne out by experiment, shows that the probability of tunneling is extremely sensitive to the value of E. For all known alpha-particle emitters, the value of E varies from about 2 to 8 megaelectron volts, or MeV (1 MeV = 10 electron volts). Thus, the value of E varies only by a factor of 4, whereas the range of is from about 10^{11} years down to about 10^{-6} second, a factor of 10^{24}. It would be difficult to account for this sensitivity of to the value of E by any theory other than quantum mechanical tunneling.

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